

McGill Baja Racing - Fundraising Announcement

Dear Friends, Family, and Alumni,

Hello, and welcome to this special edition of our newsletter! We will continue with our usual competition recap programming after our next competition in June, but in the meantime, we interrupt your day to ask you that same mildly annoying question that your kids, friends, or little siblings ask: "Hey, could you pass me ten bucks?"



Roxie (MB19) losing her wheel at EDN.

You guessed it, folks, it's time for our annual fundraising drive!

The event: McGill24

What happens: On March 11, McGill will match a portion of all donations given within a 24-hour period.

How to donate: Through [this link](#)

Is it tax-deductible: Yes (and you can get a nifty receipt!)

Why you should donate: Please please please please

Why you should *actually* donate:

This is a cool team. This is a cool project. We're students building a car that drives and jumps and flips. (That last one, we try not to.)



Jinx (MB17) flipping during a testing day.

My first week at McGill, I walked into what would become my first of many shop days and was taught how to use an angle grinder, which is a handheld rotating disc that cuts through metal and makes a big sound. Sparks were flying. I was a little scared. I tried to cut the tube, but barely made a dent. Inside the shop, a gearbox was being taken apart and cleaned. For who knows how long, I wiped old grease off the gears, marvelling at how the teeth fit together, and how every part of that machine had been designed by a student. I was hooked.



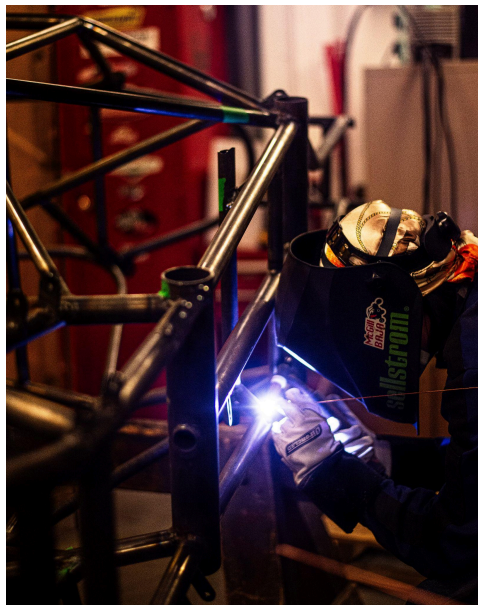
Mary (MB22) completing Hill Climb in Marana, Arizona.

Later that year, I attended the team's Internal Design Review, during which Subteam Leaders who had completed the preliminary designs for the upcoming year's car presented their ideas and received advice and critiques. The Technical Director of that year told me to write down any questions I might have, and he would answer them after the presentations. I later peppered him with questions: "What's a dog clutch?" "What's a master cylinder?" "What's the difference between an open differential and a closed one?" "What does Anti-Ackermann mean?" He answered everything. Nineteen-year-old-me forgot the replies almost immediately afterwards.



Frankie (MB23) competing in a swamp in Oshkosh, Wisconsin.

Five years later, and I know what dog clutches and master cylinders are. (A dog clutch is a small wheel with teeth perpendicular to it, which slot into a neighbouring wheel. It is used for shifting between forward, neutral, and reverse, and between two- and four-wheel drive. A master cylinder is a tube connected to the brake pedal that transfers the pedal movement into the movement of brake fluid, which in turn activates the brakes. In the car, there are two master cylinders and two dog clutches.) I have helped build multiple cars: Mary, Frankie, Kimberley, and Murphy. I have taught two new generations of welders. I have shared competition experiences with you all. And most of all, I have become a better engineer.



Author welding Kimberley (MB24)'s frame.

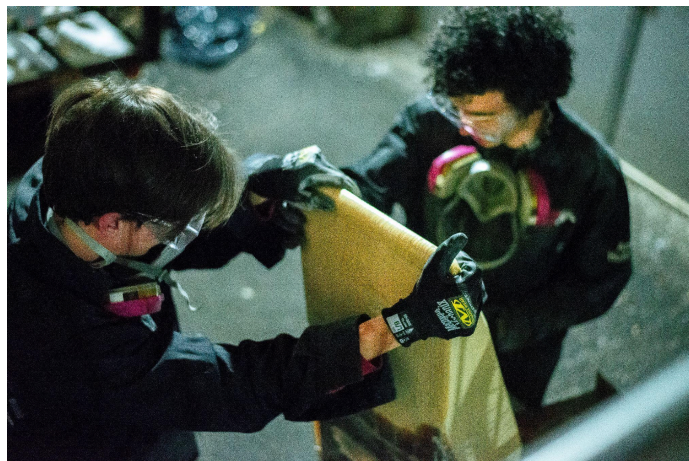
Because that's why design teams are important. Where else, as a student, are you allowed to build such a creation and then race it yourself? Where else can you be tasked with designing a part that, should it fail, could end the race, and so you learn how to build things properly? Where else will you learn to work with dozens of others to make an item that fits together so perfectly, and actually functions? Where else will you touch all the elements of the engineering

process, from the CAD ([computer-aided design](#)) to the analysis, to the purchase of steel and aluminum stock, to the communication with senior machinists and corporate sponsors, to the late-night assembly, fuelled by missed deadlines and coffee?



Teaching the next generation CADding skills.

Where else will you make such wonderful memories. Of the organized frenzy of repairs during Endurance races. Of quiet mornings when coming in to weld the frame. Of eating pizza with friends during a manufacturing break, hands still stained from grease that wouldn't come off. Of last fall's competition, watching the sun set together in a farmer's field. Of four years ago, when I successfully used the angle grinder for the first time, of watching the sparks fly and not being afraid.



Making Kimberley (MB24)'s seat mould during Shop Day.

This team doesn't just build a car. It builds lifelong skills. It builds a support system during study sessions. It builds engineers.

It builds people.

On March 11, please be generous. Your contribution will not go unnoticed.



This year's subteam leaders.

Thank you.

Yours sincerely,
Ada Santavy



Author during the -40°C cold snap at EDN. (On a lighter note: duct tape stopped sticking due to the cold, so we tried heating the roll up on a member's running car engine. (Mustang. Very cool.) Did not work.)

Thank you for getting to the end of this newsletter. If this was forwarded to you and you would like to subscribe for more updates from McGill Baja Racing, click [here](#). If you have feedback for us or no longer want to receive these messages, please reply (do not Reply All!) to this email.